NOTES FOR TMTS

Author
Wells Guan

Contents

1 Probability Theory **3**
1.1 Fundamental Conclusions in Probability 3

2 Statistical Decision Theory and Sufficiency **5**
2.1 Fundamental Definitions 5

1 Probability Theory

1.1 Fundamental Conclusions in Probability

Proposition 1.1.1 (Mills' ratio)**.** For every $t > 0$,

$$\frac{t}{t^2+1}\phi(t) \le \overline{\Phi}(t) \le \frac{\phi(t)}{t}.$$

Consequently, the logarithm of the Gaussian upper tail has leading term $-t^2/2$ as $t \to \infty$, where

$$\phi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}, \qquad \overline{\Phi}(t) = \int_t^\infty \phi(x)\,\mathrm{d}x = 1 - \Phi(t).$$

Proof. We know

$$\phi'(t) = -t\phi(t)$$

and then

$$\overline{\Phi}(t) = \int_t^\infty \phi(x)dx \le \int_t^\infty \frac{x}{t}\phi(x)dx = \frac{1}{t}\phi(x)|_\infty^t = \frac{\phi(t)}{t},$$

with

$$\overline{\Phi}(t) = \int_t^\infty \phi(x)dx = \left.\frac{\phi(x)}{x}\right|_\infty^t - \int_t^\infty \frac{\phi(x)}{x^2}dx \ge \frac{\phi(t)}{t} - \frac{\overline{\Phi}(t)}{t^2}$$

and we are done. □

Corollary 1.1.2. The moment-generating function gives the convenient, slightly looser bound

$$\mathbb{P}(|G| \ge t) \le 2e^{-t^2/2}, \qquad t \ge 0.\text{(The inequality is trivial for } t \le 1)$$

Corollary 1.1.3. Let $G_1, \ldots, G_d$ be standard normal random variables. By a union bound,

$$\mathbb{P}\left(\max_{1\le j\le d} |G_j| \ge t\right) \le 2de^{-t^2/2}.$$

Therefore, with probability at least $1-\delta$,

$$\max_{1\le j\le d} |G_j| \le \sqrt{2\log\left(\frac{2d}{\delta}\right)}.$$

The scale $\sqrt{\log d}$ is a recurring high-dimensional noise threshold.

Definiton 1.1.1 (KL Divergence)**.** Let P, Q be probability measures on $(\mathbb{R}^d, \mathcal{B}^d)$ with $P \ll Q$, and let

$$r = \frac{\mathrm{d}P}{\mathrm{d}Q}$$

be their Radon–Nikodym derivative. The Kullback–Leibler (KL) divergence is

$$D_{\mathrm{KL}}(P\|Q) = \int \log r\,\mathrm{d}P = \int r\log r\,\mathrm{d}Q,$$

where $0\log 0 = 0$. For $P \not\ll Q$, the value is defined by 0.

Definiton 1.1.2 (Mutual information)**.** The mutual information of X, Y is

$$I(X;Y) = D_{\mathrm{KL}}\big(\mu_{X,Y} \| \mu_X \otimes \mu_Y\big).$$

Definiton 1.1.3 (Conditional law and conditional mean)**.** For random vectors $X \in \mathbb{R}^p$ and $Y \in \mathbb{R}^q$, a *regular conditional distribution* of X given Y is a family $K(y, \cdot)$ of probability measures on $(\mathbb{R}^p, \mathcal{B}(\mathbb{R}^p))$ such that

$$y \longmapsto K(y, A)$$

is Borel measurable for every Borel set A, and

$$\mathbb{P}(X \in A, Y \in B) = \int_B K(y, A)\, \mu_Y(\mathrm{d}y), \qquad A \in \mathcal{B}(\mathbb{R}^p), \quad B \in \mathcal{B}(\mathbb{R}^q). \tag{1}$$

We write

$$K(y, A) = \mathbb{P}(X \in A \mid Y = y).$$

When X is real and integrable, define

$$h(y) := \int_{\mathbb{R}} x\, K(y, \mathrm{d}x)$$

where the absolute integral is finite, and set $h(y) = 0$ elsewhere. Then

$$\mathbb{E}[X \mid Y = y] = h(y) \qquad \text{and} \qquad \mathbb{E}[X \mid Y] = h(Y).$$

Proposition 1.1.4. Suppose X, Y have a joint Lebesgue density, or a joint probability mass function on a countable product support. Conditional laws are specified for μ_X-almost every x, not necessarily at every conceivable value. Then

$$I(X;Y) = \int D_{\mathrm{KL}}\big(\mu_{Y|X=x} \| \mu_Y\big)\, \mu_X(\mathrm{d}x). \tag{2}$$

Proof. Let

$$r(x, y) := \frac{\mathrm{d}\mu_{X,Y}}{\mathrm{d}(\mu_X \otimes \mu_Y)}(x, y).$$

It suffices to show that, for μ_X-almost every x,

$$\frac{\mathrm{d}\mu_{Y|X=x}}{\mathrm{d}\mu_Y}(y) = r(x, y) \qquad \text{for } \mu_Y\text{-almost every } y.$$

In the density case, on the set where $p_X(x)p_Y(y) > 0$,

$$\begin{aligned} \frac{\mathrm{d}\mu_{Y|X=x}}{\mathrm{d}\mu_Y}(y) &= \frac{p_{Y|X=x}(y)}{p_Y(y)} \\ &= \frac{p_{X,Y}(x, y)}{p_X(x)p_Y(y)} \\ &= \frac{\mathrm{d}\mu_{X,Y}}{\mathrm{d}(\mu_X \otimes \mu_Y)}(x, y). \end{aligned}$$

The same calculation applies in the discrete case, with integrals replaced by sums.

Consequently,

$$\begin{aligned}\int D_{\mathrm{KL}}\big(\mu_{Y|X=x}\|\mu_Y\big)\,\mu_X(\mathrm{d}x) &= \int_{\mathbb{R}^p}\int_{\mathbb{R}^q} r(x,y)\log r(x,y)\,\mu_Y(\mathrm{d}y)\mu_X(\mathrm{d}x)\\ &= \int_{\mathbb{R}^p\times\mathbb{R}^q} r\log r\,\mathrm{d}(\mu_X\otimes\mu_Y)\\ &= D_{\mathrm{KL}}\big(\mu_{X,Y}\|\mu_X\otimes\mu_Y\big)\\ &= I(X;Y).\end{aligned}$$

If either side is infinite, the argument is justified by applying Tonelli's theorem to

$$\psi(r) = r\log r - r + 1 \geq 0,$$

using

$$\int r\,\mathrm{d}(\mu_X\otimes\mu_Y) = 1.$$

□

2 Statistical Decision Theory and Sufficiency

2.1 Fundamental Definitions

Definiton 2.1.1. Let

$$X : (\Omega,\mathcal{F}) \longrightarrow (\mathcal{X},\mathcal{B})$$

be the random observation. Its distribution is a probability measure on the sample space $\mathcal{X}$; the realized data are a particular value x. We use P_θ for the law of the whole observation when describing a model indexed by an unknown parameter θ.

For an i.i.d. sample, write

$$X_1,\ldots,X_n \sim Q_\theta \quad \text{independently},$$

so the full sample has law

$$P_\theta = Q_\theta^{\otimes n}.$$

Here Q_θ is the **population distribution** of one unit, and n is the **sample size**. If each unit has d recorded features, then

$$x = (x_1,\ldots,x_n)^\top \in \mathbb{R}^{n\times d}$$

is a data matrix.

An observation may be a person, a measurement, or a more complicated object such as an entire graph. The statistical goal is to learn about the population, or a feature of it, from the observed sample.

Definiton 2.1.2 (Parametric model)**.** A *parametric model* is a family

$$\mathcal{P} = \big\{P_\theta : \theta\in\Theta\subseteq\mathbb{R}^k\big\}$$

with a fixed finite-dimensional parameter θ. The set Θ is the *parameter space*. Inference may concern the whole parameter or a specified function $\psi(\theta)$ of it.

Definiton 2.1.3 (Dominated model)**.** A family is *dominated* by a common sigma-finite measure ν if

$$P_\theta \ll \nu \qquad \text{for every } \theta.$$

Then its densities are

$$p_\theta = \frac{\mathrm{d}P_\theta}{\mathrm{d}\nu}.$$

Lebesgue measure gives probability density functions; counting measure gives probability mass functions. The reference measure is fixed across the family.

Definiton 2.1.4 (Identifiability)**.** A parametrization

$$\{P_\theta : \theta \in \Theta\}$$

is *identifiable* if

$$P_{\theta_1} = P_{\theta_2} \quad \Longrightarrow \quad \theta_1 = \theta_2.$$

Even in a non-identifiable parametrization, a target $\psi(\theta)$ can be identifiable: it must have the same value whenever two parameters give the same observation law.

Example 2.1.1 (Multinomial/Softmax regression)**.** A categorical response $Y \in \{1, \dots, K\}$ and observed covariates $x \in \mathbb{R}^d$, let

$$p_k(x) = P(Y = k \mid x) = \frac{\exp(\beta_k^\top x)}{\sum_{j=1}^K \exp(\beta_j^\top x)}.$$

Adding the same vector c to every β_k multiplies the numerator and denominator by $\exp(c^\top x)$, so it leaves all probabilities unchanged. Choose the last class as the reference, and set $\gamma_k = \beta_k - \beta_K$ for $k < K$ and $\gamma_K = 0$. Then

$$p_k(x) = \frac{\exp(\gamma_k^\top x)}{1 + \sum_{j=1}^{K-1} \exp(\gamma_j^\top x)}, \qquad k < K,$$

and

$$p_K(x) = \frac{1}{1 + \sum_{j=1}^{K-1} \exp(\gamma_j^\top x)}.$$

This removes the common-shift ambiguity. Identifying the coefficient vectors also requires enough variation in the design: on observed design points spanning $\mathbb{R}^d$, the log odds

$$\log\left(\frac{p_k(x)}{p_K(x)}\right) = \gamma_k^\top x$$

determine γ_k.

Definiton 2.1.5 (Statistic)**.** A *statistic* is a measurable function $T(X)$ of the observed data. The rule

$$T : \mathcal{X} \longrightarrow \mathcal{T}$$

must be specified without using an unknown parameter. Its distribution may nevertheless depend on that parameter. For real-vector data and summaries, measurability is with

respect to their Borel sigma-algebras.

Definiton 2.1.6 (Order Statistic)**.** The order statistics

$$X_{(1)} \le \cdots \le X_{(n)}$$

are the sample values sorted in increasing order, retaining repeats if there are ties. Suppose the observations are i.i.d. with CDF F and Lebesgue density f. For $1 \le i \le n$, the ith order statistic has density

$$g_i(x) = \frac{n!}{(i-1)!(n-i)!} F(x)^{i-1} \big\{1 - F(x)\big\}^{n-i} f(x), \qquad \text{for Lebesgue-a.e. } x. \tag{3}$$

Definiton 2.1.7 (Exponential family)**.** A dominated family is an *exponential family* if its densities have the form

$$p_\theta(x) = \exp\big\{\eta(\theta)^\top T(x) - \xi(\theta)\big\}\, h(x), \qquad \xi(\theta) = A\big(\eta(\theta)\big), \tag{5}$$

where $T(x) \in \mathbb{R}^m$ and $h(x) \ge 0$ do not depend on θ, and

$$A(\eta) = \log \int \exp\big\{\eta^\top T(x)\big\}\, h(x)\, \mathrm{d}\nu(x).$$

The *natural parameter* is η. Its natural parameter space $\mathcal{H}$ consists of those η for which this integral is positive and finite. In canonical form,

$$p_\eta(x) = \exp\big\{\eta^\top T(x) - A(\eta)\big\}\, h(x).$$

The function A is the *log-normalizing function*.

=

Proposition 2.1.1 (Interior differentiation)**.** On the interior $\mathcal{H}^\circ$, A is infinitely differentiable, and

$$\nabla A(\eta) = \mathbb{E}_\eta T(X), \qquad \nabla^2 A(\eta) = \mathrm{Cov}_\eta\big(T(X)\big). \tag{6}$$

For η in this interior and t sufficiently close to zero,

$$\mathbb{E}_\eta e^{t^\top T(X)} = \exp\big\{A(\eta + t) - A(\eta)\big\}.$$

Proof. By the definition of the log-normalizing function,

$$\begin{aligned}
\nabla A(\eta) &= \frac{\displaystyle\int T(x) e^{\eta^\top T(x)} h(x)\, \mathrm{d}\nu(x)}{\displaystyle\int e^{\eta^\top T(x)} h(x)\, \mathrm{d}\nu(x)} \\
&= \int T(x) e^{\eta^\top T(x) - A(\eta)} h(x)\, \mathrm{d}\nu(x) \\
&= \mathbb{E}_\eta T(X).
\end{aligned}$$

For the Hessian, differentiating the ith component of the gradient with respect to η_j gives

$$\begin{aligned}\big[\nabla^2 A(\eta)\big]_{ij} &= \partial_j \int T_i(x) e^{\eta^\top T(x) - A(\eta)} h(x)\,\mathrm{d}\nu(x) \\ &= \int T_i(x)\big(T_j(x) - \partial_j A(\eta)\big) e^{\eta^\top T(x) - A(\eta)} h(x)\,\mathrm{d}\nu(x) \\ &= \mathbb{E}_\eta[T_i(X)T_j(X)] - \mathbb{E}_\eta[T_i(X)]\mathbb{E}_\eta[T_j(X)] \\ &= \mathrm{Cov}_\eta\big(T_i(X), T_j(X)\big).\end{aligned}$$

Thus

$$\nabla^2 A(\eta) = \mathrm{Cov}_\eta\big(T(X)\big).$$

Finally, for t sufficiently close to zero so that $\eta + t \in \mathcal{H}$,

$$\begin{aligned}\mathbb{E}_\eta e^{t^\top T(X)} &= \int e^{t^\top T(x)} e^{\eta^\top T(x) - A(\eta)} h(x)\,\mathrm{d}\nu(x) \\ &= e^{-A(\eta)} \int e^{(\eta+t)^\top T(x)} h(x)\,\mathrm{d}\nu(x) \\ &= e^{A(\eta+t) - A(\eta)}.\end{aligned}$$

It remains to justify the differentiations under the integral. Since $\eta \in \mathcal{H}^\circ$, choose $\delta > 0$ small enough that $\eta + \delta s \in \mathcal{H}$ for every $s \in \{-1, 1\}^m$. For every integer $q \geq 0$, there is a finite constant $C_{q,\delta}$ such that, uniformly for $\|\eta' - \eta\|_\infty \leq \delta/2$,

$$\|T(x)\|^q e^{(\eta')^\top T(x)} h(x) \leq C_{q,\delta} \sum_{s \in \{-1,1\}^m} e^{(\eta + \delta s)^\top T(x)} h(x).$$

Every term on the right is integrable by the definition of $\mathcal{H}$. Hence dominated convergence applies to derivatives of every order, so the preceding calculations are valid and $A \in C^\infty(\mathcal{H}^\circ)$. □

Definiton 2.1.8 (Decision rule, loss, and risk)**.** A *decision rule* is a measurable map

$$\delta : \mathcal{X} \longrightarrow \mathcal{A},$$

where $\mathcal{A}$ is the action space. It turns data into an estimate, a prediction, or a testing decision. A *loss function* $L(\theta, a) \geq 0$ specifies the cost of taking action a when the parameter is θ; more generally, write $L(P, a)$. The *risk function* averages this cost over data generated under the same parameter:

$$R(\theta, \delta) = \mathbb{E}_\theta L\big(\theta, \delta(X)\big) = \int L\big(\theta, \delta(x)\big)\,\mathrm{d}P_\theta(x). \tag{7}$$

The parameter is held fixed in this expectation. Different rules give different functions of that parameter, not just one performance number. Under squared-error loss for a real target $\psi(\theta)$, the risk is its *mean squared error* (MSE):

$$R(\theta, \delta) = \mathbb{E}_\theta\left[\big(\delta(X) - \psi(\theta)\big)^2\right].$$

We use δ for a decision rule to distinguish it from a generic summary T; a deterministic

decision rule is itself a statistic.

Definiton 2.1.9 (Bias and unbiasedness)**.** For i.i.d. real observations with mean μ and finite variance σ^2, the sample mean estimates μ under squared loss and has risk

$$R(\mu, \bar{X}) = \mathbb{E}\big[(\bar{X} - \mu)^2\big] = \frac{\sigma^2}{n}.$$

For an integrable estimator $\delta(X)$ of a target $\psi(P)$, its *bias* is

$$b_\delta(P) = \mathbb{E}_P \delta(X) - \psi(P).$$

It is *unbiased* if this equals zero for every P in the model. For a square-integrable estimator,

$$\mathbb{E}_P\big[(\delta(X) - \psi(P))^2\big] = \mathrm{Var}_P\big(\delta(X)\big) + b_\delta(P)^2. \tag{9}$$

Subtract and add $\mathbb{E}_P\delta(X)$ inside the square; the cross term has expectation zero. An unbiased estimator need not have the smallest MSE. The sample variance S^2 is unbiased for σ^2, while the empirical variance with denominator n is biased.

Definiton 2.1.10 (Hypothesis testing and error probabilities)**.** Suppose $\mathcal{P}_0$ and $\mathcal{P}_1$ are disjoint and the model under consideration is $\mathcal{P}_0 \cup \mathcal{P}_1$. We test $H_0 : P \in \mathcal{P}_0$ against $H_1 : P \in \mathcal{P}_1$. A nonrandomized test takes values in $\{0, 1\}$, with 1 meaning reject H_0. Its *rejection region* is

$$C = \{x : \delta(x) = 1\},$$

and the test is the indicator $\delta(X) = 1_C(X)$, not the set C itself. Under 0–1 loss, a correct decision costs zero and an incorrect one costs one:

$$L(P, j) = \begin{cases} 0, & P \in \mathcal{P}_j, \\ 1, & P \notin \mathcal{P}_j, \end{cases} \qquad j \in \{0, 1\},$$

$$R(P, \delta) = \begin{cases} P(X \in C), & P \in \mathcal{P}_0, \\ P(X \notin C), & P \in \mathcal{P}_1. \end{cases}$$

These are the *type I error* probability under the null and the *type II error* probability under the alternative. Not rejecting a null is a decision, not a proof that it is true.

Definiton 2.1.11 (Sufficient statistic)**.** A statistic $T(X)$ is *sufficient* for $\mathcal{P} = \{P_\theta\}$ if there is a single probability kernel $K(t, B)$, not depending on θ, such that for every θ and every Borel set B,

$$P_\theta\big(X \in B \mid T(X)\big) = K\big(T(X), B\big) \qquad P_\theta\text{-almost surely.}$$

Thus the conditional distribution of the data given the statistic can be chosen to be the same across the model. Conditional laws at summary values having zero probability are interpreted through these versions, not by dividing probabilities of zero events.

Theorem 2.1.2 (Neyman–Fisher factorization)**.** Let $\{P_\theta\}$ be a family on a Euclidean Borel sample space, dominated by a common sigma-finite measure ν. A Borel statistic T with

values in a Euclidean space is sufficient if and only if its densities admit the representation

$$p_\theta(x) = g_\theta\big(T(x)\big)h(x) \qquad \nu\text{-almost everywhere for each } \theta, \tag{10}$$

where h and g_θ are nonnegative measurable functions and h does not depend on θ.